

1. Executive Summary

This project delivers an end-to-end analytical workflow designed to uncover key customer behavior patterns, retail performance drivers, and actionable opportunities for growth. Using a combination of Python, PostgreSQL, and Power BI, the analysis provides a 360° view of retail activity: customer segmentation, sales distribution, seasonal trends, and geographic performance.

The insights support data-driven decision-making in product strategy, marketing segmentation, operational planning, and revenue optimization.

2. Project Objectives

The primary goals of this analysis are:

- Identify revenue-driving product categories
 - Understand customer demographics and purchasing behaviors
 - Evaluate seasonality impact on revenue
 - Analyze geographic transaction distribution across the United States
 - Build an interactive BI dashboard consolidating KPIs and insights
 - Deliver a robust end-to-end analytical workflow as part of a professional portfolio
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3. Data Overview

The dataset consists of **3,900 customer transactions**, with detailed fields covering:

- Customer demographics (age, gender)
- Transactional data (purchase amount, previous purchases)
- Product information (category, item, size, color)

- Contextual variables (season, shipping type, discount usage)
- Behavioral signals (review rating, subscription status)
- Geographic data (U.S. state of transaction)

This multi-dimensional dataset enables a rich, business-oriented analysis across both behavioral and financial perspectives.

4. Data Preparation (Python)

Data cleaning and preparation were performed in Python using pandas and numpy.

Key steps included:

- Loading the dataset into a DataFrame
- Standardizing column names for SQL and BI compatibility
- Checking and handling missing values
- Ensuring correct data types (numeric, categorical)
- Removing duplicates
- Reviewing outliers
- Exporting the cleaned CSV for SQL analysis

The cleaned dataset was saved as:

`data/retail_customer_behavior_raw.csv`

Notebook reference:

`python/Retail_Data_Cleaning_and_Preparation.ipynb`

5. SQL Analytics (PostgreSQL)

The cleaned dataset was loaded into PostgreSQL for structured querying and metric computation.

Key analytical queries included:

5.1 Total Revenue

```
SELECT SUM(purchase_amount_usd) AS total_revenue FROM sales;
```

5.2 Total Transactions

```
SELECT COUNT(*) AS total_transactions FROM sales;
```

5.3 Average Basket Value (ABV)

```
SELECT ROUND(AVG(purchase_amount_usd::NUMERIC), 2) AS avg_basket_value FROM sales;
```

5.4 Revenue by Category

```
SELECT category, SUM(purchase_amount_usd)
FROM sales
GROUP BY category
ORDER BY SUM(purchase_amount_usd) DESC;
```

5.5 Transactions by Gender

```
SELECT gender, COUNT(*)
FROM sales
GROUP BY gender;
```

5.6 Seasonal Revenue Analysis

```
SELECT season, SUM(purchase_amount_usd)
FROM sales
GROUP BY season;
```

5.7 Geographic Distribution

```
SELECT location, COUNT(*)
FROM sales
GROUP BY location;
```

Full SQL file:

```
sql/Retail_Customer_Insights_SQL_Analysis.sql
```

The PostgreSQL analytics report is available in:

[reports/Retail_Customer_Insights_PostgreSQL_Report.pdf](#)

6. Business Intelligence Dashboard (Power BI)

An interactive Power BI dashboard consolidates the insights into a visually appealing and business-friendly format.

Dashboard file:

[powerbi/Retail_Customer_Insights_Dashboard.pbix](#)

6.1 Key KPIs

- **Total Revenue:** \$233k
- **Total Transactions:** 3,900
- **Average Basket Value:** \$59.76

6.2 DAX Measures

`Total Revenue = SUM('retail_data'[purchase_amount_usd])`

`Total Transactions = COUNTROWS('retail_data')`

`ABV = DIVIDE([Total Revenue], [Total Transactions])`

6.3 Dashboard Components

- **Revenue by Category:** Clothing is the top-performing category
- **Gender Distribution:** Male customers represent ~68% of transactions
- **Seasonality:** Spring and Fall drive the strongest revenue
- **Geographic Map:** Strong sales density across central and western U.S. states

Screenshots of the dashboard are included in the [images/](#) folder.

7. Key Insights

Based on the Python, SQL, and BI analyses, the following insights emerge:

◆ Category Performance

Clothing is the dominant revenue generator, followed by Accessories, Footwear, and Outerwear.

◆ Customer Demographics

Male customers represent the majority of purchasing activity, indicating a potential gender skew in product-market fit.

◆ Seasonality Trends

Spring and Fall show the highest revenue levels, relevant for promotional planning, inventory preparation, and marketing campaigns.

◆ Geographic Insights

State-level transaction density indicates strong performance in:

- California
- Montana
- Illinois
- Colorado
- Virginia

◆ Stable Customer Spending

ABV remains consistent at ~\$59 per order, showing stable purchasing behavior across customer segments.

8. Business Recommendations

Based on the insights, the following actions are recommended:

8.1 Product Strategy

- Increase stock and marketing investment in top-performing categories (Clothing & Accessories).

- Explore bundle opportunities to increase ABV.

8.2 Marketing Strategy

- Develop targeted campaigns for male customers (the dominant segment).
- Promote subscription programs to increase retention and recurring revenue.

8.3 Seasonal Planning

- Launch major promotions during Spring and Fall.
- Prepare inventory early to avoid stock shortages.

8.4 Geographic Strategy

- Focus marketing spend and logistics optimization in states with high transaction volumes.
- Explore regional pricing strategies where demand is strongest.

8.5 Customer Experience

- Improve low review rating segments through customer feedback loops and service enhancements.

9. Technology Stack

Layer	Technology
Data Cleaning	Python (pandas, numpy)
Data Storage	PostgreSQL
Analytics	SQL
Visualization	Power BI
Documentation & Version Control	GitHub

10. Conclusion

This end-to-end project demonstrates the complete analytical lifecycle expected from a Data Analyst or Business Analyst: cleaning, modeling, SQL analytics, visualization, insights reporting, and strategic recommendations.

It provides a strong portfolio example showcasing technical skills, business thinking, and the ability to convert raw data into actionable intelligence.